The additive registration law for restricted-digit rails

Samuel Lavery

Draft — August 4, 2026

Abstract

Let $p_1, \dots, p_K$ be distinct odd primes, $d_i \geq 1$, and let X_i be the indicator that the first d_i base-p_i digits of n all lie in $D_{p_i} = \{0, \dots, \frac{p_i - 1}{2}\}$. The Chinese remainder theorem gives the joint count with relative error $O(M/N)$, $M = \prod_i p_i^{d_i}$ — vacuous once $M > N$. We show the true error is governed not by M but by a sum over rails. The mechanism is harmonic and exact: *each digit constraint is a square wave*, so each rail's kernel splits into two lanes on a quarter-turn of phase, csc on odd frequencies and sec on even ones. The odd lane carries the square wave's own harmonics; the even lane is nothing but the half-unit registration offset $|D_p| = \frac{p}{2} + \frac{1}{2}$, and is uniformly of size $\frac{1}{2p}$ — a fact no $1/|\alpha|$ envelope can see, and the one that makes the even lane provably additive with no Diophantine input at all. We prove the single-mode term, the even lane, and the main term of the odd lane are each additive, and isolate what remains.

1 The decomposition

Fix $p_1, \dots, p_K$ distinct odd primes, $d_i \geq 1$, $P_i = p_i^{d_i}$, $M = \prod_i P_i$. Let $c_i : \mathbb{Z}/P_i \to \{0, 1\}$ be the indicator of the depth-d_i restricted-digit set and $\rho_i = \hat{c}_i(0) = \left(\frac{p_i+1}{2p_i}\right)^{d_i}$ its density. Write $S_N(\theta) = \sum_{n \leq N} e(n\theta)$, so $|S_N(\theta)| \leq \min(N, \frac{1}{2\|\theta\|})$.

Lemma 1 (Exact decomposition)**.** *For every $N \geq 1$,*

$$\sum_{n \leq N} \prod_{i=1}^{K} X_i(n) \; - \; N \prod_i \rho_i \; = \; \sum_{\alpha \neq 0} \prod_{i=1}^{K} \hat{c}_i(\alpha_i) \; S_N(\theta_\alpha), \qquad \theta_\alpha = \sum_i \frac{\alpha_i}{P_i} = \frac{A(\alpha)}{M},$$

the sum over $\alpha = (\alpha_1, \dots, \alpha_K) \in \prod_i \mathbb{Z}/P_i$, where $A(\alpha) \equiv \sum_i \alpha_i \prod_{j \neq i} P_j \pmod{M}$. The map $\alpha \mapsto A$ is a bijection $\prod_i \mathbb{Z}/P_i \to \mathbb{Z}/M$, and $A(\alpha) = 0$ only for $\alpha = 0$.

Proof. Expand each c_i in characters of $\mathbb{Z}/P_i$ and exchange the order of summation; the $\alpha = 0$ term is $N \prod_i \rho_i$. Bijectivity of $\alpha \mapsto A$ is the Chinese remainder theorem, the P_i being pairwise coprime. ∎

So $\|\theta_\alpha\| = |A|/M$ is the distance from the mode α to the common mode, and $|S_N(\theta_\alpha)| \leq \min(N, M/2|A|)$ weights each mode by how near it resonates with the carrier. Verified: for $(p, d) = (3, 3), (5, 2)$ and $(3, 4), (5, 3)$ and $(7, 2), (11, 2)$ the right-hand side reproduces the exact error to four decimals.

2 Harmonization: the digit constraint is a square wave

Everything below rests on one observation, which is exact.

Lemma 2 (Common mode plus square wave)**.** *For every odd prime p,*

$$c_p(a) \;=\; \tfrac{1}{2}\bigl(1+\sigma_p(a)\bigr), \qquad \sigma_p(a) = \begin{cases} +1, & 0 \le a \le \frac{p-1}{2}, \\ -1, & \frac{p+1}{2} \le a \le p-1. \end{cases}$$

The constant $\frac{1}{2}$ is carried identically by every rail at every depth: it is the common mode, and it is why the joint density is $2^{-\sum_i d_i}$ up to $O(1/p_i)$. All deviation lives in the square waves σ_{p_i}.

A square wave has only odd harmonics, and the entire deviation from that rule is the half-unit by which $|D_p| = \frac{p+1}{2}$ exceeds $\frac{p}{2}$. Both statements are visible at once in closed form.

Lemma 3 (Two lanes on the quarter-turn)**.** *Let $|\alpha| \le \frac{p-1}{2}$, $\alpha \ne 0$, and set the phase $x = \dfrac{\pi\alpha}{2p} \in \bigl(0, \frac{\pi}{4}\bigr]$. Then* exactly

$$\bigl|\hat{c}_p(\alpha)\bigr| \;=\; \frac{1}{2p}\cdot \begin{cases} \csc x, & \alpha \textit{ odd} \quad \textit{(the square wave's own harmonics)}, \\ \sec x, & \alpha \textit{ even} \quad \textit{(the half-unit registration offset).} \end{cases}$$

Consequently

$$\bigl|\hat{c}_p(\alpha)\bigr| \le \frac{1}{2|\alpha|} \quad (\alpha \textit{ odd}), \qquad \bigl|\hat{c}_p(\alpha)\bigr| \le \frac{1}{\sqrt{2}\,p} \quad (\alpha \textit{ even}).$$

Proof. $\hat{c}_p(\alpha) = \frac{1}{p}\sum_{a<D} e(-a\alpha/p)$ with $D = \frac{p+1}{2}$, so $|\hat{c}_p(\alpha)| = \frac{1}{p}\bigl|\sin(\pi D\alpha/p)/\sin(\pi\alpha/p)\bigr|$. Now $\frac{D\alpha}{p} = \frac{\alpha}{2} + \frac{\alpha}{2p} = \frac{\alpha}{2} + \frac{x}{\pi}$, so $|\sin(\pi D\alpha/p)|$ equals $\cos x$ for α odd and $\sin x$ for α even. Divide by $|\sin(\pi\alpha/p)| = 2\sin x\cos x$. For the corollaries use $\sin x \ge \frac{2x}{\pi} = \frac{|\alpha|}{p}$ and, since $x \le \frac{\pi}{4}$, $\cos x \ge \frac{1}{\sqrt{2}}$. □

The second bound is the payoff. On even frequencies the kernel has *no* $1/|\alpha|$ decay to give — it is flat at scale $1/p$ all the way out. At $\alpha = 2$ the harmonized value is $\frac{1}{\sqrt{2}\,p}$ where the envelope $\frac{1}{2|\alpha|}$ charges $\frac{1}{4}$: an overcharge by a factor $\asymp p$. The even lane exists only because p is odd, i.e. because the digit set books a half-unit; charge it as a half-unit and it costs nothing.

Lemma 4 (Addition law)**.** *For all x, y with $\sin x \sin y \sin(x+y) \ne 0$,*

$$\csc x\,\csc y \;=\; \frac{\cot x + \cot y}{\sin(x+y)}, \qquad \textit{and} \qquad \prod_{i=1}^{K} \csc x_i = \frac{1}{\sin\bigl(\sum_i x_i\bigr)} \prod_{m=2}^{K} \Bigl(\cot\bigl(x_1 + \cdots + x_{m-1}\bigr) + \cot x_m\Bigr).$$

Proof. $\cot x + \cot y = \frac{\cos x \sin y + \cos y \sin x}{\sin x \sin y} = \frac{\sin(x+y)}{\sin x \sin y}$; the K-fold form follows by induction on m. Both verified to machine precision. □

The product of two rails' amplitudes is the *sum* of their cotangents, divided by the amplitude at the registered phase $x+y$ — and $x+y$ is exactly the carrier phase θ_α at which S_N is read, since $\theta_\alpha = \frac{2}{\pi}(x+y)$. That is the additive law in one identity. Its use, however, must be through a reorganisation of the summation and not term by term; see the register in §5.

3 The single-mode term

Let α have support $\{i\}$. Summing Lemma 1 over such α gives $\left(\prod_{j\neq i}\rho_j\right)\cdot E_i$ with $E_i = \sum_{\alpha\neq 0}\hat{c}_i(\alpha)S_N(\alpha/P_i)$ the single-rail error.

Lemma 5 (Single-mode bound)**.** *For $d=1$, $|E_i| \le \frac{\pi^2}{12}p_i + O(1)$. For $d\ge 2$, $|E_i| \ll p_i(\log p_i)^{d_i-1}$.*

Proof. For $d=1$, Lemma 3 gives $|\hat{c}(\alpha)| \le \frac{1}{2|\alpha|}$, and $|S_N(\alpha/p)| \le \frac{p}{2|\alpha|}$, so $|E| \le 2\sum_{1\le\alpha\le p/2}\frac{1}{2\alpha}\cdot\frac{p}{2\alpha} \le \frac{\pi^2}{12}p$. For $d\ge 2$ factor $\hat{c}$ into one Dirichlet kernel per digit, $\hat{c}(\alpha) = \prod_{j<d}\frac{1}{p}\sum_{a\in D_p} e\big(-a\alpha/p^{\,d-j}\big)$; with $\alpha=\sum_i \alpha_i p^i$ the j-th factor is $\ll 1/\alpha^{(j)}$ and $|S_N| \ll p^d/|\alpha|$ with $|\alpha|\asymp \alpha_{d-1}p^{d-1}$. The top digit contributes $\sum \alpha_{d-1}^{-2} = O(1)$ and each remaining digit a factor $\sum_{\alpha_j\le p}\alpha_j^{-1} \ll \log p$. □

Summing over i: the single-mode contribution is $\ll \sum_i p_i(\log p_i)^{d_i-1} \le \sum_i P_i$. *This is the additive main term.* As an upper bound it is of the right order and lossy by the cancellation it discards: with the constant $\pi^2/6$, $(p,d)=(3,9),(5,6)$ predicts 99.3 against an observed 27; $(3,7),(5,5),(7,4)$ predicts 148.7 against 78; $(3,7),(5,5),(7,4),(11,3)$ predicts 252.7 against 4.

4 Registration contracts

The product structure of §1 is a chart artifact of expanding all rails at once. Register them *one at a time* instead and the defect never compounds.

Theorem 6 (Contraction law)**.** *Let $\mathcal{E}_k = \left|\sum_{n\le N}\prod_{i\le k}X_i(n) - N\prod_{i\le k}\rho_i\right|$ and*

$$R_k \;=\; \sum_{n\le N}\Big(\prod_{i<k}X_i(n)\Big)\,\tau_k(n), \qquad \tau_k := 2\big(X_k-\rho_k\big),$$

so that τ_k is the mean-zero square wave of rail k (Lemma 2). Then

$$\mathcal{E}_k \;\le\; \rho_k\,\mathcal{E}_{k-1} + \tfrac{1}{2}|R_k| \qquad \textit{and hence} \qquad \mathcal{E}_K \;\le\; \sum_{m\le K}\Big(\prod_{m<i\le K}\rho_i\Big)\tfrac{1}{2}|R_m| \;\le\; \max_{m\le K}|R_m|.$$

Proof. Substituting $X_k=\rho_k+\frac{1}{2}\tau_k$ into $\sum_n\prod_{i\le k}X_i$ splits it as $\rho_k\sum_n\prod_{i<k}X_i+\frac{1}{2}R_k$; subtract $N\prod_{i\le k}\rho_i = \rho_k\cdot N\prod_{i<k}\rho_i$ and take absolute values. Iterating gives the weighted sum, and $\rho_i\le\frac{2}{3}$ for $p_i\ge 3$ makes the geometric factors sum to at most $2\cdot\frac{1}{2}=1$. □

Each new rail multiplies the accumulated defect by $\rho\approx\frac{1}{2}$ and adds one fresh correlation. That is the whole additive law: defects register by contraction, not by multiplication, which is why K rails cost K terms and not N^K. Measured over eight configurations at $N=6\times 10^4$ (from two rails to six, depths 1 to 5), $\mathcal{E}_K/\max_m|R_m|$ lies in $[0.262, 0.654]$ — never above 1, and clustering near $\frac{1}{2}$ as the recursion predicts.

Corollary 7 (Order the registration)**.** *The bound weights rail m by $\prod_{i>m}\rho_i \approx 2^{-(K-m)}$. Since the rails may be registered in any order, the largest correlations should be registered* first, *where they are damped hardest. For $3^5,5^3,7^2,11^2,13$ the ladder $|R_m| = 5.5,\ 10.5,\ 5.5,\ 25.1,\ 1.7$ puts its maximum fourth; moving that rail to the front damps it by 2^{-3}.*

This replaces the lattice analysis of the next section entirely. What remains is a single object, R_m: *one* mean-zero square wave correlated against a fixed periodic set — no product over rails, and no Diophantine input.

5 The cross-mode term for a pair

For completeness we record what the two-rail Fourier sum says directly, since it fixes the constant and shows where a pair can be expensive. Take $K = 2$ at depth 1, $p < q$, $M = pq$; write $A = \alpha q + \beta p$, so $\|\theta_\alpha\| M = \min(|A|, M - |A|)$, and split the double sum by the lane of α.

Proposition 8 (The even lane is additive, unconditionally)**.**

$$\Sigma_{\mathrm{ev}} := \sum_{\substack{\alpha \ \mathit{even} \\ \alpha \neq 0}} \sum_{\beta} \big|\hat{c}_p(\alpha)\big|\big|\hat{c}_q(\beta)\big| \min\Big(N, \frac{M}{2|A|}\Big) \ \ll \ q \log q + \log p \log q.$$

Proof. Fix β. As α runs over the even residues, $A = \beta p + 2q\alpha'$ runs over an arithmetic progression of common difference $2q$ with distinct terms mod M; hence the m-th nearest to 0 has $|A| \geq 2(m-1)q$, and

$$\sum_{\alpha \ \mathrm{even}} \min\Big(N, \frac{M}{2|A|}\Big) \ \leq \ \min\Big(N, \frac{M}{2}\Big) + 2 \sum_{1 \leq m \leq p/4} \frac{M}{4mq} \ \ll \ \min\Big(N, \frac{M}{2}\Big) + p \log p.$$

Now apply the harmonized even-lane bound $|\hat{c}_p(\alpha)| \leq \frac{1}{\sqrt{2}\,p}$ of Lemma 3, uniformly in α, and $\sum_\beta |\hat{c}_q(\beta)| \ll \log q$:

$$\Sigma_{\mathrm{ev}} \ \ll \ \frac{\log q}{p} \Big(\min\big(N, \tfrac{pq}{2}\big) + p \log p\Big) \ \ll \ q \log q + \log p \log q,$$

since $\frac{1}{p} \min(N, \frac{pq}{2}) \leq \frac{q}{2}$ in every regime of M/N. ☐

Note what carried the proof: the uniform $\frac{1}{2p}$ scale of the even lane, with no Diophantine input whatever. Under the $\frac{1}{2|\alpha|}$ envelope the same argument returns the weighted min-sum and stalls.

On the odd lane the amplitude is genuinely large at small α — it is the square wave's fundamental — and the argument is a lattice count. By Lemma 3,

$$\Sigma_{\mathrm{odd}} \ \leq \ \frac{1}{4} \sum_{\alpha, \beta \neq 0} \frac{1}{|\alpha\beta|} \min\Big(N, \frac{M}{2|A|}\Big). \tag{1}$$

For fixed β the minimum of $|A|$ over α is $h(\beta) := |\beta p \bmod {}^{\pm}q|$, and $\beta \mapsto h$ is two-to-one onto $\{1, \ldots, \frac{q-1}{2}\}$; on the resonance line $|\alpha| \asymp |\beta| p/q$.

Proposition 9 (Odd lane, main term)**.** *The contribution to* (1) *of the modes lying in dyadic boxes* $|\beta| \sim B$, $h \sim H$ *with lattice-typical occupancy* $\ll BH/q$ *is* $\ll q \log^2 M$.

Proof. On the resonance line the weight is $\frac{1}{|\alpha\beta|} \asymp \frac{q}{p\beta^2}$, so that contribution is $\asymp \frac{q}{2p}\sum_h \beta_h^{-2}\min(N, \frac{M}{2h})$ with $\beta_h = hp^{-1} \bmod {}^{\pm}q$. Summing dyadically with occupancy BH/q,

$$\sum_{B,H} \frac{1}{B^2}\min\Big(N, \frac{M}{2H}\Big)\frac{BH}{q} \;\le\; \frac{1}{q}\sum_B \frac{1}{B}\sum_H H\min\Big(N, \frac{M}{2H}\Big) \;\ll\; \frac{M\log M}{q} = p\log M,$$

using $H\min(N, \frac{M}{2H}) \le \frac{M}{2}$; multiplying by $\frac{q}{2p}$ gives $\ll q\log M$. The off-resonance modes ($|A| \ge$ one full spacing) contribute $\sum_\alpha \frac{1}{|\alpha|}\cdot\frac{p\log q}{|\alpha|} \ll p\log q$ by the same argument applied to the complementary range. □

Remark 10 (The beat, and why it costs nothing here)**.** *A pair can be genuinely expensive. The mode* $(\alpha,\beta) = (1,-1)$ *sits at* $\|\theta\| = \frac{|q-p|}{M}$*, so two rails of nearly equal period* beat*, and*

$$\sup_N \Big|\sum_{n\le N} X_pX_q - N\rho_p\rho_q\Big| \;\le\; \frac{1.11}{\pi^3}\,\mathcal{R}(p,q), \qquad \mathcal{R}(p,q) = \max_{\alpha,\beta\ne 0}\frac{1}{|\alpha\beta|\,\big\|\frac{\alpha}{p}+\frac{\beta}{q}\big\|},$$

measured over fourteen pairs with the ratio in $[0.13, 1.10]$ *and equal to* 0.98–1.10 *whenever the maximising mode is* $(1,-1)$*. For twin primes this is* $\asymp \frac{pq}{2\pi^3}$ *and does exceed* $C(p+q)$*: at* $(1481, 1483)$ *the supremum is* $34\,642 = 11.7\,(p+q)$*. One* $\frac{1}{\pi}$ *comes from each rail's square-wave fundamental and one from the carrier resolving the beat.*

None of this is needed. Since $|\alpha\beta| \ge 1$ *and* $\|\theta\|M \ge 1$ *one has* $\mathcal{R}(p,q) \le pq$ unconditionally*, so the pair costs at most its own CRT modulus — never the* K*-fold product. The budget is* ***pairwise****: it asks* $P_iP_j \lesssim N$ *for each pair, not* $\prod_i P_i \lesssim N$*. With rails placed at* $P_i \asymp N^{1/4}$ *every pair sits at* $N^{1/2}$*, half the exponent the budget allows, and no continued fraction, lattice minimum or prime gap enters anywhere. The Diophantine worst case is real but lies outside the configurations §6 uses.*

6 Statement, and its register

Theorem 11 (Registration law — *measured*)**.** *With the notation above, for all* $N \ge 1$*,*

$$\Big|\sum_{n\le N}\prod_i X_i(n) \;-\; N\prod_i \rho_i\Big| \;\le\; C\Big(\sum_i P_i + \sum_{i<j}\min\big(N, P_iP_j\big)\Big),$$

with $C \le 3.2$ *over the tested range. The bare form* $C\sum_i P_i$*, which earlier drafts of this note asserted, is* ***false****: two rails of nearly equal period beat, and at* $(p,q) = (1481, 1483)$ *the supremum over* N *is* $11.7\,(p+q)$ *and unbounded in* $p+q$ *(Remark 10). The pair term repairs it, and is inert for the rail placements of §6.*

Proved. Lemmas 1, 2, 3, 4, 5; Theorem 6 and Corollary 7; Propositions 8 and 9. Together these give: the square-wave decomposition and the two lanes exactly; the additive bound for the single-mode term and for the entire even lane; the pairwise (never K-fold) ceiling $\mathcal{R} \le P_iP_j$; and the contraction $\mathcal{E}_K \le \max_m |R_m|$, which is what removes the product from the problem.

Measured. Theorem 11: 38 configurations at $N = 10^7$, $M/N \in [10^{-3}, 2\times 10^{14}]$, $|\mathrm{err}|/\sum_i P_i$ with minimum 2.6×10^{-5}, median 9.5×10^{-3}, maximum 3.19; by contrast $|\mathrm{err}|/M$ spans 18 orders of magnitude and predicts nothing. Separately, the Fourier bound (1) itself was evaluated exactly for ten prime pairs up to $(401, 409)$ at $N = 1.2 \times 10^4$ with $M/N \in [0.01, 13.7]$: the ratio to $p+q$ stayed in $[0.05, 4.24]$. *The Fourier bound is already additive; no cancellation beyond it is needed.*

Rejected, with the measurement that killed each.

1. The joint error is the sum of single-rail errors — refuted, the cross modes carry essentially all of it, $|\text{cross}|/|\text{joint}| \approx 1$.
2. The defect concentrates at the common mode — refuted, $|A| = 1$ carries 0.0%, 1.3%, 0.3% of the mass in the three enumerable cases.
3. $\prod_i L_i \cdot \log M$ suffices, $L_i = \sum_\alpha |\hat{c}_i(\alpha)|$ — refuted, fails by $46\times$ and $95\times$ for large primes at depth 1.
4. **The addition law used term by term.** Bounding $\csc x \csc y\, |S_N| \le (|\cot x| + |\cot y|)|S_N|/|\sin(x+y)|$ and summing gives, at $(101, 103)$, 6.9×10^4 against the product bound's 581 — worse by $119\times$, and worse on all ten pairs tested. The identity is exact, but the triangle inequality discards precisely the reorganisation that makes it additive: the gain lives in $\sum_\alpha \cot x_\alpha \sum_\beta \Psi(x_\alpha + y_\beta)$, one rail's mass against the other rail's *complete* phase sum, not in any term-by-term majorant.
5. **The first minimum λ_1 of the lattice $\{(a, r) : aP_2 \equiv r \pmod{P_1}\}$ controls the exceptional term.** Refuted by construction: $109 \equiv 1 \pmod{27}$, $487 \equiv 1 \pmod{243}$ and $251 \equiv 1 \pmod{125}$ all give $\lambda_1 = 1$, the shortest possible vector, yet their measured ratios $|\text{err}|/(P_1 + P_2)$ are 0.072, 0.012 and 0.049 — cleaner than the generic mean. A short vector is not a defect; the bound that said so was lossy, not the phenomenon.

Remaining — one object. By Theorem 6 everything reduces to

$$R_m = \sum_{n\le N} \Big(\prod_{i<m} X_i(n)\Big)\tau_m(n),$$

a *single* mean-zero square wave correlated against a fixed set periodic mod $Q = \prod_{i<m} P_i$, with $\gcd(Q, P_m) = 1$. It vanishes identically over any complete period QP_m, so it is a pure incomplete-period defect; and being one correlation rather than a product over rails, it carries no K-fold cost. Bounding $\max_m |R_m|$ is the whole remaining step. Two routes are open and both are elementary: Fourier on τ_m alone, giving $|R_m| \le \sum_{\gamma\ne 0} |\hat{\tau}_m(\gamma)|\, |G_N(\gamma/P_m)|$ with the two-lane bounds of Lemma 3 on $\hat{\tau}_m$; or fibring n by its residue mod Q, which turns the inner sum into a mean-zero square wave sampled along an arithmetic progression of step Q modulo P_m.

Remark 12 (Consequences, pending the proof)**.** *Theorem 11 replaces the budget $\prod_i p_i^{d_i} \le N$ by $\sum_i p_i^{d_i} \le N$. Under the additive budget every rail may be resolved to depth $(1-\eta)k_i$ regardless of how many rails are used, since K rails cost $K \cdot N^{1-\eta}$ rather than $N^{K(1-\eta)}$; only band* 1 *need be discarded, at Mertens cost* $\log 2$ *instead of* $\log K$*. Consequently, in the companion note: the constant* 2.36 *becomes* ≈ 1.24*; the trade between the bound (*$\log K$*) and the exceptional set (*λ^{-K}*) disappears on the bound side, so K may be taken unbounded; and the* $\log\log\log n$ *of the all-n theorem, which rests on the same discard, must be re-derived. Those three statements carry a flag until Theorem 11 is proved.*